SmartCare – Smart Hospital Management System

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Abstract

SmartCare is a web-based hospital management system designed to organise common hospital administration activities through a single application. In many healthcare settings, patient registration, doctor information, appointments, consultation records, pharmacy, laboratory services, admissions and billing may be managed through separate processes. This can make information difficult to coordinate and may lead to repeated data entry or delays in accessing records. The project aims to bring these activities together and support a more organised workflow for authorised users. The system is developed using Java 17, Jakarta Servlets and JSP, JDBC and MySQL, with Maven used for project packaging. Its described modules include authentication and administration, patient and doctor management, appointment and consultation handling, pharmacy, laboratory, nursing, billing, admissions, reports and feedback. Password protection and role-based access are important design considerations for limiting access to system functions. The project can be evaluated through functional, integration, usability, security and performance testing. Such testing should verify that records are stored consistently, workflows behave as expected, and users receive clear feedback when an action succeeds or fails. This document describes the system as an academic software project; its features and security should not be considered fully validated until the relevant tests have been completed and documented.

Keywords: Smart Hospital Management System, Hospital Information System, Java, JSP, Servlets, JDBC, MySQL, Patient Management, Appointment Scheduling, Data Security.